

Definition 6.2. A *inner product* of a vector space V is a function that maps $u, v \in V$ to $\langle u, v \rangle \in F$, and it satisfies:

- *Positivity:* $\langle v, v \rangle \geq 0$.
- *Definiteness:* $\langle v, v \rangle = 0 \iff v = 0$.
- *Additivity:* $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
- *Homogeneity:* $\langle \lambda v, w \rangle = \lambda \langle v, w \rangle$.
- *Conjugate Symmetry:* $\langle u, v \rangle = \overline{\langle v, u \rangle}$

Definition 6.4. A vector space equipped with an inner product is called an *inner product space*.

We assume vector spaces V, W are inner product space for the rest of chapter.

Theorem 6.6. *Properties of inner product:*

- Let $v \in V$, then $\langle -, v \rangle$ is a linear map $V \rightarrow F$.
- For any $v \in V$, we have $\langle 0, v \rangle = \langle v, 0 \rangle = 0$.
- For any $u, v, w \in V$, $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$.
- For any $v, w \in V$ and $\lambda \in F$, $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$.

Proof.

- Trivial from the definition.
- First, $\langle 0, v \rangle = 0$ since $\langle -, v \rangle$ is a linear map, thus maps 0 to 0. Then $\langle v, 0 \rangle = \overline{\langle 0, v \rangle} = \overline{0} = 0$.
- $\langle u, v + w \rangle = \overline{\langle v + w, u \rangle} = \overline{\langle v, u \rangle + \langle w, u \rangle} = \overline{\langle v, u \rangle} + \overline{\langle w, u \rangle} = \langle u, v \rangle + \langle u, w \rangle$.
- $\langle u, \lambda v \rangle = \overline{\langle \lambda v, u \rangle} = \overline{\lambda \langle v, u \rangle} = \overline{\lambda} \overline{\langle v, u \rangle} = \bar{\lambda} \langle u, v \rangle$

□

Definition 6.7. For any $v \in V$, the **norm** of v is denoted by $\|v\|$, and is defined by:

$$\|v\| = \sqrt{\langle v, v \rangle}$$

Theorem 6.9. Let $v \in V$,

- $\|v\| = 0 \iff v = 0$.
- For any $\lambda \in F$, $\|\lambda v\| = |\lambda| \|v\|$.

Proof.

- Trivial by the definition of inner product.
- $\|\lambda v\| = \langle \lambda v, \lambda v \rangle = \sqrt{\lambda \bar{\lambda}} \langle v, v \rangle = |\lambda| \|v\|$.

□

Definition 6.10. Let $u, v \in V$, u and v are **orthogonal** $\iff \langle u, v \rangle = 0$

Theorem 6.11.

- 0 is orthogonal to any $v \in V$.
- 0 is the only vector that orthogonal to itself.

Proof. Both trivial by the definition, (1) is equivalent to $\langle 0, v \rangle = 0$ and (2) is equivalent to $\langle v, v \rangle = 0 \iff v = 0$. □

Theorem 6.12 (勾股定理). Let $u, v \in V$, if u is orthogonal to v , then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Proof.

$$\begin{aligned} \|u + w\|^2 &= \langle u + w, u + w \rangle \\ &= \langle u, u + w \rangle + \langle w, u + w \rangle \\ &= \langle u, u \rangle + \langle u, w \rangle + \langle w, u \rangle + \langle w, w \rangle \\ &= \langle u, u \rangle + \langle w, w \rangle \end{aligned}$$

The last equation is by $\langle u, w \rangle = \langle w, u \rangle = 0$ cause u, w are orthogonal. □

Theorem 6.13 (One Orthogonal Factorization). Let $u, v \in V$ and $v \neq 0$.

Let $c = \frac{\langle u, v \rangle}{\|v\|^2}$ and $w = u - cv$, then $u = cv + w$ and $w \perp v$.

Proof. 嗯算。 □